

BUSINESS ANALYTICS – WEEK 6 TASK

Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions

Task Objective: Analyze 15 million user-path logs and use a Markov Chain model to measure the contribution of each marketing channel to conversions.

1. Problem Statement

Students will ingest 15 million user path logs containing sequences of marketing channel touches. The objective is to construct a transition probability matrix, calculate the probability of conversion, and determine the Removal Effect of every channel by removing that channel from the transition graph and measuring the relative reduction in conversion probability.

2. Input Data

Field	Example	Description
User_ID	U10025	Unique user identifier
Timestamp	2026-09-01 10:15	Time of interaction
Channel	Google Ads	Marketing channel
Path	Social → Email → Search	Sequence of channel touches
Conversion	1	1 = converted, 0 = not converted

3. Business Analytics Methodology

Step 1 – Data Ingestion: Load the 15 million user-path records using a scalable processing system.

Step 2 – Data Cleaning: Remove duplicate paths, handle missing channels, standardize channel names, and sort touches by timestamp.

Step 3 – Path Construction: Represent each customer journey as a sequence such as Start → Social → Email → Search → Conversion.

Step 4 – Transition Counting: Count how many times each channel transitions to another channel or to Conversion/Null.

Step 5 – Transition Probability Matrix: Divide each transition count by the total outgoing transitions from that channel.

Step 6 – Baseline Conversion Probability: Calculate the probability that a customer eventually reaches Conversion from the Start state.

Step 7 – Removal Effect: Remove one channel at a time, rebuild the transition relationships, and recalculate conversion probability.

Step 8 – Ranking: Rank channels according to their Removal Effect and identify the most influential marketing channels.

4. Markov Chain Transition Matrix

Let P be the transition probability matrix. Each element P_{ij} represents the probability of moving from state i to state j . For a channel i :

$$P_{ij} = \text{Number of transitions from } i \text{ to } j / \text{Total outgoing transitions from } i$$

Example Matrix

From / To	Social	Email	Search	Conversion	Null
Social	0.00	0.40	0.30	0.20	0.10
Email	0.10	0.00	0.50	0.30	0.10
Search	0.10	0.20	0.00	0.60	0.10

5. Removal Effect Calculation

First calculate the baseline conversion probability $P(\text{Conversion})$. Then remove a selected channel and calculate the new conversion probability $P_{-i}(\text{Conversion})$. The Removal Effect for channel i is:

$$\text{Removal Effect}(i) = [P(\text{Conversion}) - P_{-i}(\text{Conversion})] / P(\text{Conversion})$$

A higher Removal Effect means that the channel has a larger estimated contribution to the conversion journey.

6. Small Worked Example

Assume the baseline probability of conversion is 0.50. After removing the Email channel, the conversion probability falls to 0.35.

$$\text{Removal Effect}(\text{Email}) = (0.50 - 0.35) / 0.50 = 0.30 = 30\%$$

Therefore, Email has a 30% Removal Effect in this example. Repeat the same procedure for every channel.

7. Suggested Implementation Workflow

15 Million Logs

- Data Cleaning
- Customer Journey Paths
- Transition Counts
- Transition Probability Matrix
- Baseline Conversion Probability
- Remove Each Channel
- Recalculate Conversion Probability
- Removal Effect
- Channel Ranking & Business Insight

8. Expected Business Insights

- Identify the marketing channels that contribute most strongly to conversion journeys.
- Detect channels that appear frequently but have low incremental contribution.
- Compare channel importance using a consistent mathematical attribution method.
- Support budget allocation and campaign optimization decisions.

- Use the results to improve multi-channel marketing strategy.

9. Tools / Technologies

Recommended: Python, Pandas, NumPy, SQL, Apache Spark/PySpark for large-scale processing, and Power BI or Tableau for visualization. For 15 million records, distributed processing is preferable when the available system cannot efficiently handle the data in memory.

10. Deliverables

- Cleaned and processed user-path dataset.
- Transition count table.
- Transition probability matrix.
- Baseline conversion probability.
- Removal Effect value for every marketing channel.
- Ranked channel-attribution table.
- Visualization of channel Removal Effects.
- Final business recommendations.

11. Conclusion

The Markov Chain attribution method provides a mathematical way to evaluate how marketing channels influence customer conversion paths. By systematically removing each channel and measuring the reduction in conversion probability, the business can identify high-impact channels and make better marketing investment decisions.